Managing IT Projects:

Risk, Quality & Integration

IT3120 – Modern Enterprise Management

Waterfall / **Agile** / Hybrid / AI-Assisted Project Management

Where Are We in Project Management?

BUILDING ON WHAT WE KNOW

Project lifecycle and process groups

Stakeholder management

Communication management

Scope management

Cost management

Schedule management

Resource management

Procurement management

TODAY'S FOCUS

Risk Management

Quality Management

Integration Management

Managing change

Waterfall vs Agile approaches

AI-assisted project management



Learning Outcomes

BY THE END OF THIS LECTURE, YOU WILL BE ABLE TO:

Explain risk, quality and integration management in IT projects

Identify and evaluate common IT project risks

Develop suitable responses to project risks

Define and measure quality for IT deliverables

Explain how integration and change control coordinate a project

Compare these practices under Waterfall, Agile and Hybrid approaches

Use Generative AI critically as a project-management assistant



Why Are IT Projects Difficult to Manage?

Significant uncertainty drives IT projects

Requirements, technology, and threats keep changing

Complex integrations and hidden dependencies

Progress and quality are hard to see

Defects and new opportunities surface late

User adoption remains unpredictable

Can a project manager
simply follow the original plan?



What Does IT Project Success Mean?

TRADITIONAL VIEW

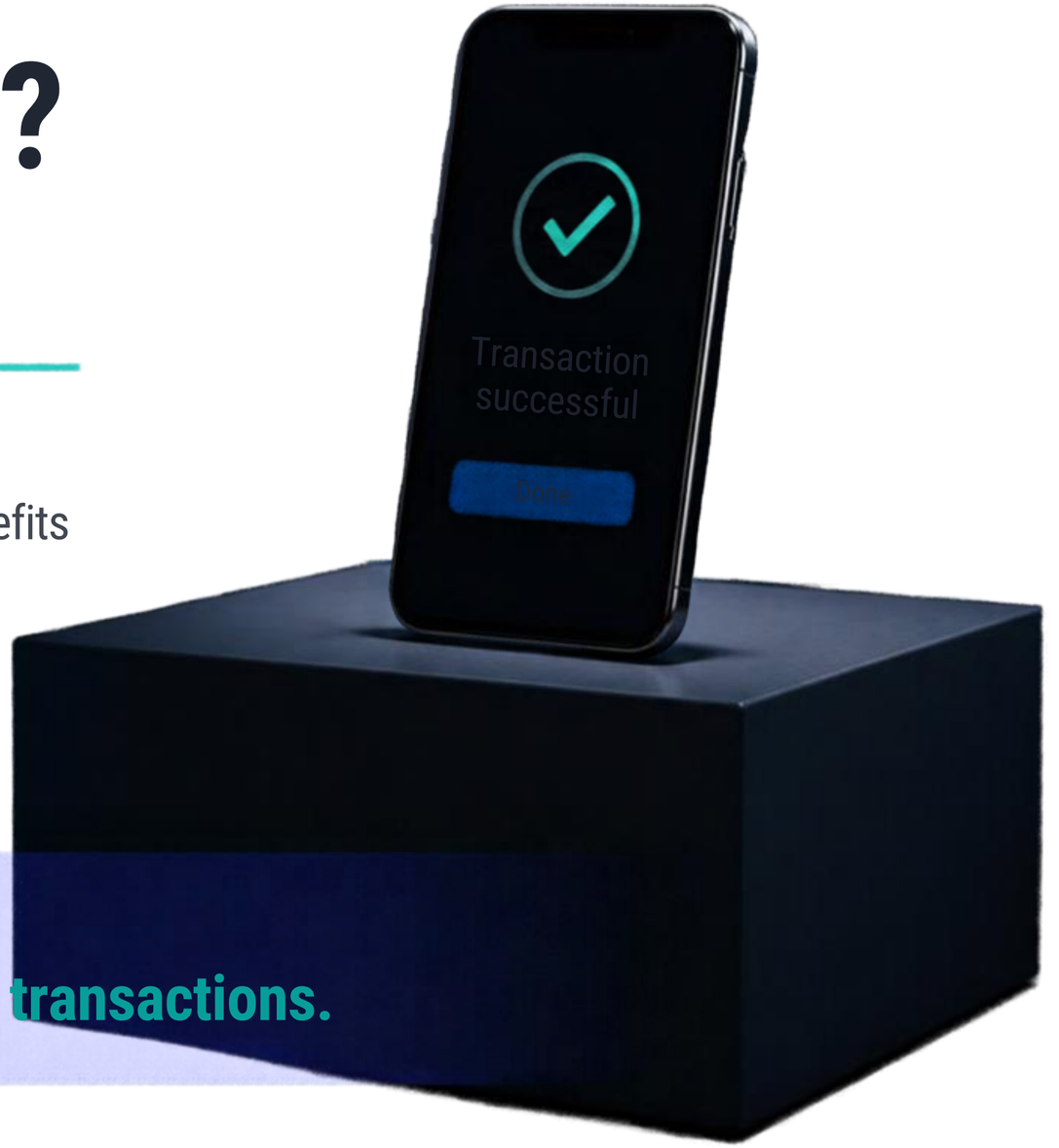
Scope + Time + Cost

An IT project may meet all three and still fail.

MODERN VIEW

Scope + Time + Cost + Quality + Risk + User Value + Business Benefits

Success is about outcomes that create real value.



A mobile app delivered on time and within budget
is not successful if users cannot reliably complete transactions.

The Three Questions for Today's Lecture

RISK

What could happen?

QUALITY

How good must the solution be?

INTEGRATION

How do we keep everything working together?

These three management areas are highly interconnected.

One Project Management Method Does Not Fit Every IT Project

Different projects call for different approaches.

Predictive / Waterfall. Adaptive / Agile. Hybrid.

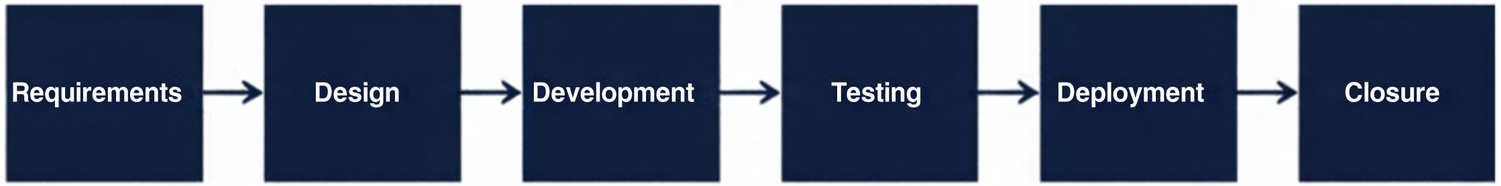
The management principles remain important.

How they are applied changes.



Predictive / Waterfall Approach

The traditional, plan-driven way IT projects are delivered.



Work is planned in detail before execution begins

Scope is defined early and remains relatively stable

Formal documentation and baselines guide the project

Changes follow a formal control process

Testing activities are concentrated later in the cycle

Best suited when requirements can be reasonably defined upfront



Example

University ERP Implementation

1. Requirements
2. System specification
3. Procurement
4. Configuration/ development
5. Data migration
6. Testing
7. User training
8. Deployment
9. Acceptance

Changes may have major effects on:

- Cost
- Schedule
- Contracts
- Testing



Agile Approach

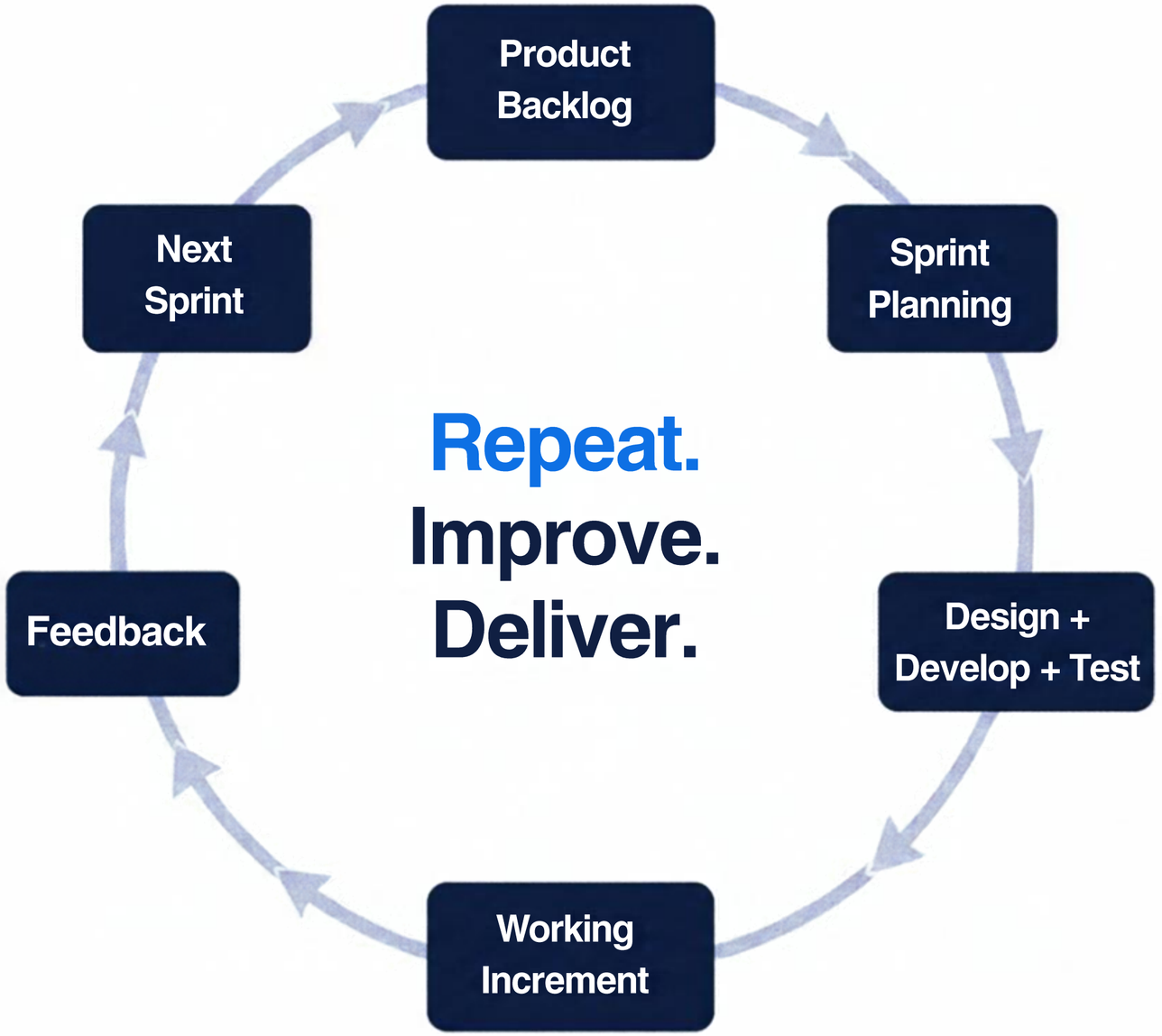
Agile delivers value through short, **iterative** cycles.

Work happens in focused sprints

We build, review, and adapt

Feedback drives continuous improvement

Value is delivered incrementally



Key Agile Concepts for Managers

ROLES

Product Owner
Prioritizes value and backlog.

Scrum Master
Facilitates the process.

Development Team
Produces the increment.

ARTIFACTS & PRACTICES

- Product Backlog
- User Stories
- Sprint Backlog
- Definition of Done
- Increment
- Sprint Review
- Retropective



Example User Story

As a registered user, I want to **reset my forgotten password** so that I can regain access to my account.

ACCEPTANCE CRITERIA

- OTP sent to registered email mobile
- OTP expires after specified time
- Password policy enforced
- User notified after successful change

Quality starts with clear requirements.



Hybrid IT Project Management

Real IT projects succeed by using the right approach for each part of the work.

PREDICTIVE	AGILE	DEVOPS
Budget	Software modules	Testing
Procurement	UI development	Deployment
Infrastructure	Reports	Monitoring
Major milestones	Dashboards	

A hybrid approach is not a compromise.
It is choosing the best practices for each part of the project.



Waterfall vs Agile

Issue	Predictive/Waterfall	Agile
Requirements	Defined early	Evolve
Planning	Upfront	Iterative
Scope	Baseline	Product backlog
Change	Change request	Backlog reprioritization
Delivery	Major releases	Increments
Testing	Defined test phases	Continuous
Customer feedback	Milestones	Frequent
Ris k	Formal reviews/register	Continuous



CLASS ASSIGNMENT 1

AI-Assisted Activity: Select the Approach

7
minutes

Groups of
3–4

Goal
Compare, evaluate, decide

Your group must

Identify two good points in the AI response

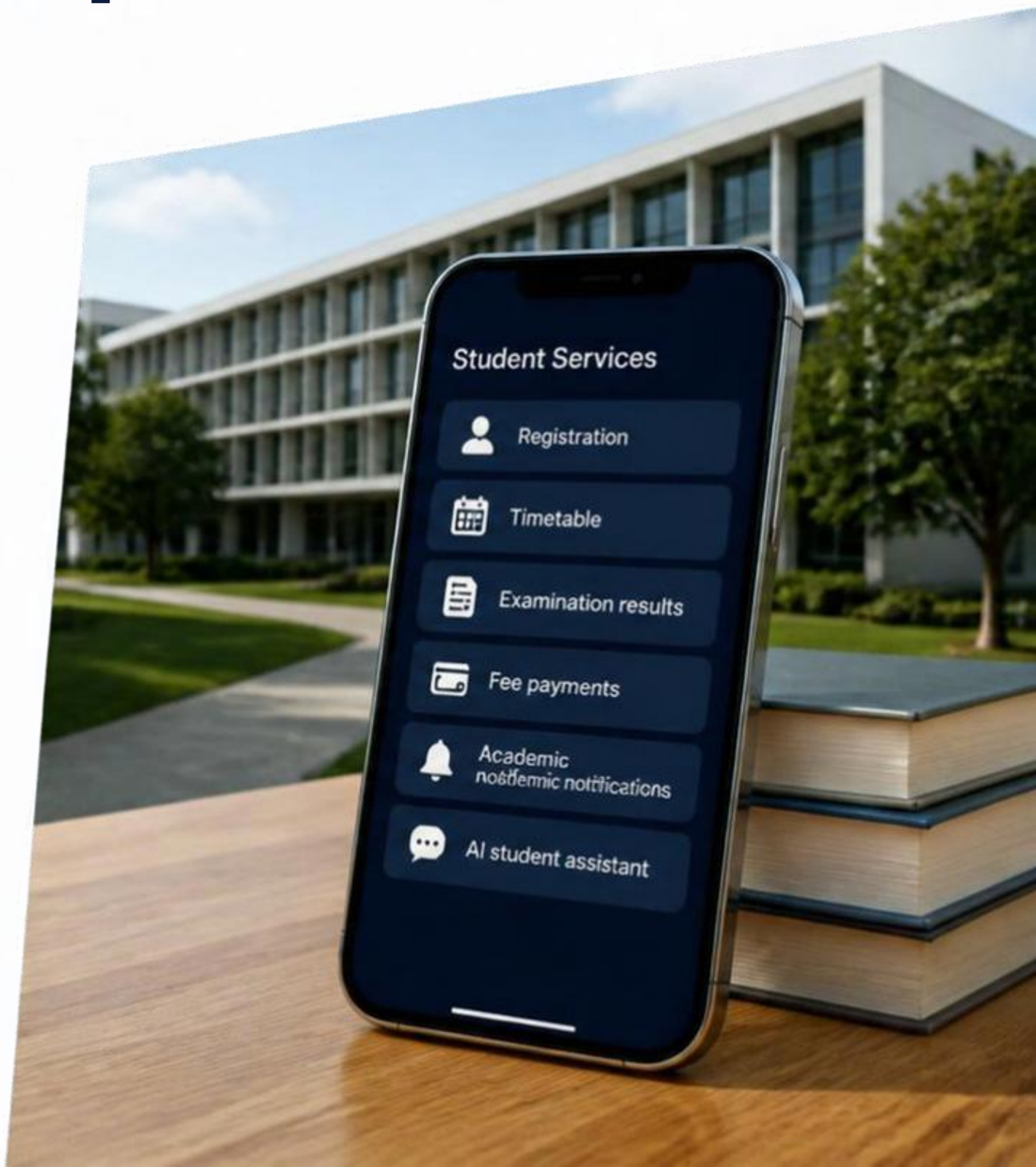
Identify one questionable assumption

Decide your group's preferred approach

Give two reasons

Submit

AI recommendation + your group's decision



WHAT IS PROJECT RISK?

An uncertain event or condition that, if it occurs, can have a positive or negative effect on project objectives.



THREAT

Negative effect

=

OPPORTUNITY

Positive effect



Effective project risk management treats both **threats** and **opportunities**.

Risk Is NOT the Same as an Issue

Risk

Something that may happen.

Example:

The cloud provider may experience an outage during registration.

Issue

Something that has happened.

Example:

The cloud service is currently unavailable.

Assumption =
Something believed to be true

Constraint =
A limitation placed on the project

COMMON IT PROJECT RISK CATEGORIES



Requirements

Unclear/changing requirements



Technical

New technology, API failure, legacy integration



Security

Cyberattack, data leakage



People

Key developer leaves



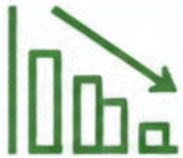
Schedule

Development takes longer



Vendor

Supplier/APIcloud failure



Business

Poor user adoption



External

Regulatory or technology changes

Technical, schedule, security, scope and external risks are common in IT projects.

RISK MANAGEMENT CYCLE



Risk management is continuous, not a one-time exercise.

FORMAL RISK MANAGEMENT PROCESSES

PMBOK-oriented processes:

1. Plan Risk Management
2. Identify Risks
3. Perform Qualitative Risk Analysis
4. Perform Quantitative Risk Analysis
5. Plan Risk Responses
6. Implement Risk Responses
7. Monitor Risks



Students should understand the workflow, rather than simply memorize seven names.

IDENTIFYING IT PROJECT RISKS

Techniques include:

- Brainstorming
- Interviews
- Expert judgment
- Checklists
- SWOT
- Root Cause Analysis
- Lessons learned
- Assumption analysis
- AI-assisted brainstorming



AI can suggest risks. AI does not know your project automatically.

WRITING A GOOD RISK STATEMENT

Avoid:

Cybersecurity

Better:

If authentication vulnerabilities remain undetected,
then unauthorized users could access student information,
resulting in privacy, reputational and operational impacts.

CAUSE → **RISK EVENT** → **EFFECT**

Risk Register

ID	Risk	P	I	Rating	Response	Owner
R01	API unavailable	3	5	15	Mitigate	Tech Lead
R02	Key developer leaves	2	4	8	Mitigate	PM
R03	Requirements change	4	4	16	Mitigate	Product Owner

The risk register is a living management document.

Qualitative Risk Analysis

How likely is it?

Probability

How serious would it be?

Impact

Example scale:

1 = Very Low	2 = Low	3 = Medium	4 = High	5 = Very High

Risk Score = Probability × Impact

Probability–Impact Matrix

Probability ↓ / Impact →	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

Should every risk receive equal management attention?

No—prioritize.

Quantitative Risk Analysis

For significant projects we may quantify uncertainty.

Examples:

- Expected Monetary Value

  EMV = Probability × Financial Impact

Other techniques:

- Decision tree analysis
- Sensitivity analysis
- Monte Carlo simulation

For IT3120, students mainly need to understand when and why they are useful.

Responding to Threats



AVOID

Remove the cause.



MITIGATE

Reduce probability and/or impact.



TRANSFER

Shift responsibility/financial exposure.



ACCEPT

Take no preventive action beyond monitoring/contingency.

IT Examples of Threat Responses

Risk:	New untested database technology
Avoid →	Use established technology.
Mitigate →	Prototype and performance-test it.
Transfer →	Specialist vendor assumes defined responsibility.
Accept →	Proceed and maintain contingency resources.

The response depends on project context.

Opportunities Are Risks Too



Positive uncertainty can create value.

Strategies:

- Exploit
- Enhance
- Share
- Accept

Example: A new AI API could reduce development effort by 30%.

Management question: Should the project exploit the opportunity?

Consider benefits and new risks.

Risk Management in Agile

Agile does not eliminate risk management.

Risk reduction occurs through:

- Short iterations
- Early prototypes
- Technical spikes
- Frequent demonstrations
- Continuous testing
- Backlog refinement
- Sprint reviews
- Retrospectives

Fail early → Learn early → Correct early

CLASS ASSIGNMENT 2

AI-Assisted Risk Register

Time: 8 minutes

Scenario

A team is developing an online examination system for 10,000 university students.

Features:

- Login
- Question bank
- Timed examinations
- Automatic submission
- Online marking
- Results
- Cloud hosting

Use AI to:

Generate 10 possible project risks.

Then your group must:

1. Select the five most important.
2. Rewrite them using If → Then format.
3. Assign probability (1–5).
4. Assign impact (1–5).
5. Calculate risk score.
6. Select a response.
7. Assign a risk owner.

Which AI-generated risk did you reject, and why?

What Is Quality?



Quality is not:

“The software has many features.”

Quality concerns the degree to which requirements and stakeholder needs are fulfilled.

Quality ≠ Grade

A simple application can be high quality.

A feature-rich application can be poor quality.

The original presentation correctly emphasizes meeting requirements and prevention rather than simply inspecting defects later.

What Does Quality Mean in Software?

Consider:

- Functionality
- Reliability
- Usability
- Performance
- Security
- Availability
- Maintainability
- Compatibility
- Scalability
- Accessibility



Which is the most important?

It depends on the system and stakeholder requirements.

Example: Different Systems, Different Quality Priorities

Mobile Game

- Performance
- Usability
- Availability

Banking App

- Security
- Reliability
- Accuracy
- Availability

University LMS

- Usability
- Scalability
- Availability

Hospital Information System

- Reliability
- Security
- Availability
- Data integrity

Quality must therefore be planned.

Three Quality Management Processes

PLAN QUALITY

What quality is required?

↓

MANAGE QUALITY

Are appropriate processes being followed?

↓

CONTROL QUALITY

Does the product actually meet requirements?

Quality Assurance vs Quality Control

Quality Assurance – QA

Process focused.

Are we doing the work correctly?

Examples:

- Development standards
- Code-review process
- QA audits
- CI practices

Quality Control – QC

Product focused.

Is the output correct?

Examples:

- Testing
- inspection
- defect analysis
- UAT

Prevention Is Better Than Correction

Poor approach:

Develop → Find hundreds of defects → Repair

Better:

Requirements review

Design review

Coding standards

Code review

Automated testing

Integration testing

UAT

Cost of Quality

Cost of Conformance

- Prevention
- Training
- Reviews
- Testing

versus

Cost of Non-Conformance

- Rework
- Service failure
- Customer complaints
- Security incidents
- Lost customers
- Reputation damage

Question: Is testing a cost or an investment?

IT Quality Metrics



Quality should be measurable.

Examples:

- Response time < 2 seconds
- Availability ≥ 99.9%
- Critical defects = 0 before release
- Failed transactions < 0.1%
- Test pass rate ≥ 98%
- Security vulnerabilities by severity
- Customer reported defects
- Mean Time to Restore



Avoid:

“System must be fast.”



Prefer:

“95% of requests must respond within 2 seconds.”

Testing Is Part of Quality Management



Examples:

- Unit testing
- Integration testing
- System testing
- Regression testing
- Performance testing
- Security testing
- Usability testing
- User Acceptance Testing

Different tests address different risks.

Quality in Agile



Quality should not wait until the end of the project.

Agile practices include:

- Acceptance criteria
- Definition of Done
- Code review
- Automated testing
- Continuous Integration
- Frequent demonstrations
- Retrospectives

The original presentation already identifies unit/integration testing, code reviews, CI/CD, Definition of Done, UAT and security/performance testing as IT quality practices.

Definition of Done



Example:

A user story is DONE when:

- ✓ Code completed
- ✓ Peer reviewed
- ✓ Unit tests passed
- ✓ Integration tests passed
- ✓ Security checks passed
- ✓ Documentation updated
- ✓ Acceptance criteria satisfied
- ✓ Product Owner accepts it

Developer finished coding Done.

CLASS ASSIGNMENT 3

AI-Assisted Quality Planning

Time: 8 minutes

Scenario

Your team is developing a mobile banking app supporting:

- Fund transfer
- Bill payment
- Account balances
- Biometric login

Ask AI: "Suggest measurable quality requirements and quality metrics for this mobile banking application."

Your group must:

1. Select five useful metrics.
2. Reject at least one poor/vague metric.
3. Improve one AI-generated metric so it becomes measurable.
4. Develop a Definition of Done for "Transfer Funds".
5. Identify three types of testing required.

Be prepared to explain why your requirements are appropriate.

Part E – Project Integration Management

What Is Project Integration Management?

An IT project contains many interconnected parts:



But these cannot be managed independently.



Integration Management asks:

How do we coordinate all parts of the project so that they work together to achieve the business objective?

The existing lecture describes integration as combining and coordinating project activities, managing interdependencies, making trade-offs and keeping the project aligned with business goals.

Integration requires:

- Coordination
- Trade-offs
- Decision-making
- Change management
- Knowledge sharing
- Maintaining alignment with business objectives

Why Is Integration Especially Important in IT Projects?

Consider a seemingly simple request:



"Add biometric login to our mobile application."





This is not merely a scope change.



One change can affect the entire project.

That is why integration management is needed.

Impacts across the project:

Scope	→	New functionality
Schedule	→	Additional development/testing
Cost	→	Additional effort/licensing
Resources	→	Security expertise
Quality	→	Reliability/security requirements
Risk	→	Authentication/privacy risks
Procurement	→	Third-party API/SDK
Stakeholders	→	Users/security/compliance

Integration Is the ‘Big Picture’

A project manager should not ask only:

“Can the developers build it?”



The PM must ask:

- Should we build it?
- What business benefit does it provide?
- What else will change?
- What new risks will appear?
- What will it cost?
- Will the schedule change?
- Do we have the required skills?
- What testing is required?
- Who must approve it?
- Is it still worth doing?

Integration = making decisions across the whole project.

Seven Integration Management Processes

The traditional PM framework identifies:

1. **Develop Project Charter**
2. **Develop Project Management Plan**
3. **Direct and Manage Project Work**
4. **Manage Project Knowledge**
5. **Monitor and Control Project Work**
6. **Perform Integrated Change Control**
7. **Close Project or Phase**



Think of these as:

START → PLAN → DO → LEARN → CHECK → CHANGE → CLOSE